EXPERIMENT 2

ntp_server.py Python

```python
import ntplib
from datetime import datetime
import time


def synchronize_clock(ntp_server="pool.ntp.org"):

    client = ntplib.NTPClient()

    try:
        print(f"Connecting to NTP Server: {ntp_server}\n")

        response = client.request(ntp_server, version=3, timeout=5)
        # Local system time
        system_time = time.time()

        # Time received from NTP server
        ntp_time = response.tx_time

        system_readable = datetime.fromtimestamp(system_time)
        ntp_readable = datetime.fromtimestamp(ntp_time)

        print("Local System Time :", system_readable)
        print("NTP Server Time   :", ntp_readable)

        # Calculate clock offset
        offset = ntp_time - system_time

        print("\nClock Offset:", round(offset, 6), "seconds")

        if abs(offset) < 1:
            print("System clock is almost synchronized.")
        else:
            print("System clock differs from NTP server.")

    except Exception as e:
        print("Error while connecting to NTP server:", e)


if __name__ == "__main__":
    synchronize_clock("time.google.com")
```

Output:

PROBLEMS DEBUG CONSOLE TERMINAL PORTS OUTPUT zsh - expt_2

```
(venv) ➜  expt_2 git:(main) ✗ pip install ntplib
Requirement already satisfied: ntplib in ./venv/lib/python3.9/site-packages (0.4.0)
WARNING: You are using pip version 21.2.4; however, version 26.0.1 is available.
You should consider upgrading via the '/Users/shaileshsathe/Projects/dist_comp/expt_2/venv/bin/python3 -m
 pip install --upgrade pip' command.
(venv) ➜  expt_2 git:(main) ✗ python3 ntp_server.py
Connecting to NTP Server: time.google.com

Local System Time : 2026-03-09 14:42:36.796013
NTP Server Time   : 2026-03-09 14:42:36.802786

Clock Offset: 0.006773 seconds
System clock is almost synchronized.
(venv) ➜  expt_2 git:(main) ✗
```